KPK BOARD TEXTBOOK NOTES

Physics - Class 9

Unit 7: Properties of Matter

Complete notes in English, based on *A Textbook of Physics, Grade IX*
Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar

> **How to use these notes.** Read one section at a time, learn the green definition boxes by heart, and copy every yellow formula box into your notebook. Work through the solved numericals before you try the multiple-choice questions (MCQs) at the end. The last page is a one-page revision sheet for the night before the exam.

What this unit covers

1. Kinetic molecular model of matter - solids, liquids and gases, and plasma, the fourth state
2. Density - mass per unit volume
3. Pressure - force per unit area, and everyday examples
4. Atmospheric pressure, the barometer, altitude and weather
5. Pressure in liquids - P = ρhg and its properties
6. Pascal's law and hydraulic machines
7. Archimedes' principle, upthrust and the principle of floatation
8. Elasticity, Hooke's law, stress, strain and Young's modulus

After studying this unit, you should be able to

- State the kinetic molecular model of matter (solid, liquid and gas forms) and describe briefly the fourth state of matter, plasma.
- Define density, and compare the densities of a few solids, liquids and gases.
- Define pressure as a force acting normally on unit area, and explain how pressure varies with force and area in everyday examples.
- Explain that the atmosphere exerts a pressure, describe how the height of a liquid column may be used to measure it, and state that it decreases with height above the Earth's surface.
- Explain that changes in atmospheric pressure in a region may indicate a change in the weather; state Pascal's law with examples, and use P = ρgh for pressure beneath a liquid surface.
- State Archimedes' principle and determine the density of an object using it; state the upthrust exerted by a liquid on a body and the principle of floatation.
- Define stress, strain and Young's modulus, state Hooke's law and explain the elastic limit.

This checklist follows the Student Learning Outcomes list printed at the start of Unit 7 in the textbook (book page 171).

7.1 Kinetic Molecular Model of Matter

> **Kinetic molecular model:** all matter is made of tiny particles called molecules, which are in constant motion. The strength of the forces between the molecules and the freedom with which they can move decide whether the matter is a solid, a liquid or a gas.

The textbook opens the unit with a simple question: what is common in an ice cube and water? The answer is that both are the same substance - water - in two different states. Ice, water and steam look and behave differently only because their molecules are arranged and move differently.

Feature	Solid	Liquid	Gas
Molecular arrangement	Molecules are packed closely in a regular pattern	Molecules are close, but not in a regular pattern	Molecules are far apart
Intermolecular forces	Very strong	Weaker than in solids	Very weak, almost negligible
Motion of molecules	Vibrate about fixed positions only	Can slide and move past one another	Move freely and rapidly in all directions
Shape	Fixed shape	Takes the shape of the container	Fills the whole container
Volume	Fixed volume	Fixed volume	No fixed volume
Examples	Ice, iron, wood	Water, oil, milk	Air, steam, oxygen

Plasma - the fourth state of matter

At very high temperatures, the atoms of a gas lose their electrons and the gas becomes a mixture of free electrons and positively charged ions. This highly ionised state is called **plasma**. It is the fourth state of matter.

- Plasma is found in the Sun and other stars, where temperatures are extremely high.
- On Earth it appears in lightning, in neon and fluorescent tubes, and in plasma displays.
- Most of the visible matter in the universe exists as plasma.

> **Fluids:** liquids and gases are collectively called fluids, because both can flow. Almost everything in this unit after density - pressure, Pascal's law and Archimedes' principle - is about fluids.

7.2 Density

Density is mass per unit volume. (Textbook, Key Points)

$$\text{Density } \rho = \text{mass} / \text{volume} \quad \text{or} \quad \rho = m / V$$

The SI unit of density is the kilogram per cubic metre, **kg m^{-3}**. Density tells us how tightly the matter of a substance is packed. For the same volume, the substance with the greater density has the greater mass: a brick feels heavier than a same-sized block of thermocol because brick is denser.

Densities of some common substances

Substance	Density (kg m^{-3})	Substance	Density (kg m^{-3})
Gold	19,300	Water (pure)	1,000
Mercury	13,600	Ice	920
Lead	11,300	Oil (typical)	about 900
Silver	10,500	Wood (typical)	about 700
Copper	8,900	Oxygen (gas)	1.43
Iron / steel	7,800	Air	1.29
Aluminium	2,700	Hydrogen (gas)	0.09

Read the table from top to bottom and three facts stand out. Solids are generally denser than liquids, and liquids are far denser than gases - a gas has roughly one-thousandth the density of its liquid. Ice (920 kg m^{-3}) is less dense than water (1,000 kg m^{-3}), which is why ice floats on water. Mercury is so dense that even iron can float on it.

Solved example - mass from density

Question (from the book's Exercise MCQ 1). A container having volume 6 m^3 is full of a liquid of density 30 kg m^{-3}. Find the mass of the liquid.

Solution. From $\rho = m / V$, we get $m = \rho V$ = 30 kg m^{-3} x 6 m^3 = **180 kg**.

Measuring density in the laboratory

- **Regular solid:** measure its mass on a balance, find its volume from its dimensions, then use $\rho = m / V$.
- **Irregular solid:** find its volume by the displacement of water in a measuring cylinder.
- **Liquid:** the textbook's lab work finds the density of a liquid using a 5 ml syringe instead of a density bottle - weigh the syringe empty, fill it with a known volume of the liquid, weigh it again, and divide the mass of the liquid by its volume.

7.3 Pressure

Pressure is force applied per unit area - the force acting normally (at right angles) on unit area of a surface. (Textbook, Key Points)

$$\text{Pressure } P = F / A \quad \text{Unit: pascal (Pa)} = 1 \text{ N m}^{-2}$$

Pressure depends on two things only: the size of the force and the area over which it acts.

- For a **fixed area**, a larger force gives a larger pressure.
- For a **fixed force**, a smaller area gives a larger pressure.

Everyday examples from the textbook

- **Finger and needle on a balloon.** Press a finger against a balloon and it only makes a small indentation. Push a needle with the same force and the balloon bursts - the same force on the tiny area of the needle tip produces enough pressure to rupture it.
- **Heeled shoes.** Being stepped on by a heeled shoe hurts more than by a flat one, because the same body weight acts on a much smaller area. Spiked shoes are painful for the same reason: the weight is concentrated on a small area.
- **Sharp knife.** The cutting edge of a knife is made very thin so that a small force produces a large pressure and cuts easily.
- **Camel's feet.** A camel's broad feet spread its weight over a large area, so the pressure on the sand is small and it can walk in the desert without sinking.
- **Wide straps and foundations.** School bags have wide shoulder straps, and buildings have wide foundations, to reduce pressure.

Solved Example 7.2 - Weight as pressure on the ground (textbook)

Question. A 40-kg girl's two feet cover an area of 500 cm^2. (a) Determine the pressure exerted by the two feet on the ground. (b) If the girl stands on one foot, what will be the pressure under that foot?

Given. Force F = weight W = mg = 40 kg x 9.8 m s^{-2} = 392 N. Area A = 500 cm^2 = 500 x 10^{-4} m^2 = 0.05 m^2.

(a) P = F / A = 392 N / 0.05 m^2 = 7,840 N m^{-2} = 7.8 x 10^3 N m^{-2} = **7.8 kPa**.

(b) On one foot the force is unchanged but the area is halved, so the pressure doubles: P = **15.6 kPa**.

Try the book's extension: what is the pressure if she stands with all her weight on a heel tip of area 0.0001 m^2? P = 392 / 0.0001 = 3,920,000 Pa = 3.92 MPa - about 500 times the two-feet value.

7.4 Atmospheric Pressure

> **Atmospheric pressure** is the pressure that the atmosphere exerts. We live at the bottom of an ocean of air, and the air pushes inward on our bodies just as water pushes on a swimmer. (Textbook, 7.4)

A fluid exerts pressure in every direction. Swimmers and divers feel water pressure on all parts of their bodies, and we are under air pressure all the time - we do not notice it because the pressure inside our bodies balances it. Pump the air out of a sealed can and the unopposed atmospheric pressure crushes the can inwards, which shows how large this pressure really is.

$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} \quad | \quad 1 \text{ bar} = 1.000 \times 10^5 \text{ Pa}$$

- At sea level, atmospheric pressure is on average **1.013×10^5 N m^{-2}** (1.013×10^5 Pa). This value defines the unit called the **atmosphere (atm)**.
- The **bar** is the unit used in meteorology and on weather maps: 1 bar = 1.000×10^5 Pa, so standard atmospheric pressure is slightly more than 1 bar.
- The pressure of the air at a given place varies slightly according to the weather.

The barometer

> **Barometer:** a device used to measure atmospheric pressure. (Textbook, Key Points)

In a mercury barometer, atmospheric pressure pushes on the mercury in an open dish and supports a column of mercury in an inverted glass tube. The height of the column therefore measures the pressure: at sea level the column stands about **76 cm (760 mm)** high. Because pressure in a liquid is $P = \rho hg$, measuring the height h of the liquid column is a way of measuring atmospheric pressure.

Pressure, altitude and weather

- **Altitude:** atmospheric pressure **decreases** with height above the Earth's surface, because the air becomes thinner. Climbers at high altitude face lower pressure and thinner air, which causes breathing difficulties due to the lower level of oxygen. The mercury column of a barometer carried up a mountain falls as you climb.
- **Ears popping:** while driving in the mountains, the outside pressure changes faster than the air inside the ear can adjust, so the ears 'pop' as the pressures equalise.
- **Weather:** changes in atmospheric pressure in a region may indicate a change in the weather - a falling barometer often warns of rain or a storm, while a rising barometer usually means fair weather.

Pressure in Liquids

Consider a liquid of density ρ standing in a container. At a depth h below the surface, the liquid above a small area A forms a column of volume V = Ah. Its mass is m = ρV = ρAh, so its weight - the force on the area - is F = mg = ρAhg. Dividing by the area gives the textbook's equation 7.4:

$$P = F / A = \rho A h g / A \quad \text{so} \quad P = \rho\, h\, g \quad \text{(Equation 7.4)}$$

Pressure in a liquid therefore depends on only three things: the **density** of the liquid, the **depth** h, and the acceleration due to gravity g. It does not depend on the shape of the container or the total amount of liquid. Points A, B, C and D at the same distance h beneath the surface of a liquid in connected vessels of different shapes are all at the same pressure, and the liquid stands at the same level in every vessel.

Properties of pressure in liquids (textbook)

1. **Pressure increases with depth,** because the further down you go, the greater the weight of liquid above. In a can with holes at different heights, water spurts out fastest and furthest from the lowest hole.
2. **At one depth, pressure acts equally in all directions.** A can with similar holes all round it at the same level spurts water equally fast and equally far from every hole.
3. **Pressure depends on density:** for the same depth, a denser liquid (for example mercury) exerts a greater pressure than a less dense one (for example water).

Solved example - pressure under water

Question. Find the pressure due to water at a depth of 2 m. (Density of water ρ = 1,000 kg m^{-3}, g = 9.8 m s^{-2}.)

Solution. P = ρhg = 1,000 x 2 x 9.8 = **19,600 Pa**. This is the pressure from the water alone; the total pressure at that depth also includes the atmospheric pressure acting on the surface.

Why this matters

- **Dams** are built thicker at the bottom, where the water pressure is greatest.
- **Water tanks** are placed at the highest level of a house so that the depth h is large and the water reaches every tap with good pressure.
- **Divers** must be careful at depth, where the pressure on the body becomes very large.

7.5 Pascal's Law and Hydraulic Machines

> **Pascal's principle:** if the pressure at one point of a confined fluid is increased by an amount, the pressure increases by the same amount at all other parts throughout the fluid. (Textbook, Key Points)

In short: pressure applied to an enclosed fluid is transmitted equally in all directions, without any loss. A fluid cannot be compressed easily, so a push at one end is felt in full at the other end - this is what makes hydraulic machines possible.

The hydraulic lift

Two pistons of different areas are connected by a fluid-filled pipe. A small force F_1 on the small piston of area A_1 creates a pressure $P = F_1 / A_1$. By Pascal's law the same pressure acts under the large piston of area A_2, producing a much larger force F_2:

$$F_1 / A_1 = F_2 / A_2 \quad \text{so} \quad F_2 = F_1 \times (A_2 / A_1)$$

The force is multiplied by the ratio of the piston areas. The trade-off is distance: the small piston must move down much further than the large piston rises, so no energy is gained - only force.

Applications of Pascal's law

- **Hydraulic lift and jack** - a small effort lifts a car for repair or washing.
- **Hydraulic brakes** - pressure from the brake pedal is transmitted through brake fluid equally to all four wheels.
- **Hydraulic press** - compresses scrap metal and shapes metal parts.
- **Excavators and loaders** - hydraulic rams move their heavy arms.

Solved Numerical 3 - hydraulic lift (textbook, page 196)

Question. The small piston of a hydraulic lift has an area of 0.20 m^2. A car weighing 1.20 x 10^4 N sits on a rack mounted on the large piston, which has an area of 0.90 m^2. How large a force must be applied to the small piston to support the car?

Solution. By Pascal's law, $F_1 / A_1 = F_2 / A_2$, so $F_1 = F_2 \times A_1 / A_2$ = 1.20 x 10^4 N x 0.20 / 0.90 = **2.67 x 10^3 N** (about 2,667 N). A force of under 3,000 N supports a car weighing 12,000 N.

7.6 Archimedes' Principle, Upthrust and Floatation

When a body is placed in a fluid, the pressure at the bottom of the body is greater than at the top, because pressure increases with depth. The difference pushes the body upwards. This upward force is called the **buoyant force** or **upthrust**, and it makes the body appear to lose weight.

> **Archimedes' principle:** a body, when completely or partially immersed in a fluid, will experience an upthrust equal to the weight of the fluid displaced by the body. This upthrust brings about an apparent loss of weight of the body. (Textbook, Key Points)

Upthrust = weight in air - apparent weight in fluid = weight of fluid displaced

Principle of floatation

> **Principle of floatation:** a floating body displaces its own weight of the fluid in which it floats.

- If the **weight of the body is greater** than the upthrust, the body **sinks** - a small needle of steel sinks because its density is greater than water's.
- If the **weight equals** the maximum upthrust, the body **floats** - a huge steel ship floats because its hollow shape displaces a large volume of water, so it displaces its own weight before it is fully submerged.
- A body floats higher in a denser liquid, and a submarine rises or sinks by filling or emptying its ballast tanks to change its overall density.

Finding density using Archimedes' principle

Weigh a solid in air (W_1) and then fully immersed in water (W_2). The loss of weight $W_1 - W_2$ equals the weight of water displaced, which gives the volume of the solid. Then density = mass / volume.

Solved Numerical 5 - immersed block (textbook, page 196)

Question. A block is fully immersed in water. Before immersion it weighed 30 N; while immersed, its apparent weight was 25 N. Calculate (a) the upthrust, (b) the weight of water displaced, (c) the volume of water displaced, (d) the volume of the block, and (e) the density of the block.

Solution. (a) Upthrust = 30 - 25 = **5 N**. (b) By Archimedes' principle, weight of water displaced = upthrust = **5 N**. (c) Weight = ρVg, so V = 5 / (1,000 x 9.8) = **5.1×10^{-4} m^3**. (d) The block is fully immersed, so the volume of the block = volume displaced = **5.1×10^{-4} m^3**. (e) Mass of block = 30 / 9.8 = 3.06 kg, so density = 3.06 / 5.1×10^{-4} = **6,000 kg m^{-3}**.

7.9 Elasticity and Hooke's Law

> **Elasticity:** the property of solid materials to return to their original shape and size after the removal of the deforming force. (Textbook, 7.9)

A rubber band returns to its original length when the distorting force is removed, an archer's bow springs back after the arrow is released, and a tennis ball regains its shape after bouncing off the racket. Materials that do not return to their original shape are **inelastic** - plasticine, clay and dough.

> **Elastic limit:** most materials are elastic only up to a certain limit, called the elastic limit. Beyond the elastic limit, a material will not return to its original dimensions - it breaks or is permanently deformed when the deforming force is removed.

7.9.1 Hooke's law

> **Hooke's law:** within elastic limits, the extension (or compression) is directly proportional to the force applied. (Textbook, Key Points)

$$F_{res} \propto -x \quad \text{or} \quad F_{res} = -k\,x \quad \text{(Equation 7.6)}$$

Here k is the **force constant** (also called the spring constant), with unit **N m^{-1}**. The negative sign shows that the restoring force is directed against the displacement. The same relationship holds for a wire under tension. A stiff spring has a large k; a soft spring has a small k.

7.10 Stress, Strain and Young's Modulus

Quantity	Definition (textbook)	Formula and unit
Stress	The force applied per unit area of cross-section to produce deformation	Stress = F / A; unit N m^{-2} or pascal (Pa)
Strain	The extension per unit length	Strain = x / l; **no unit** - it is a ratio of two lengths
Young's modulus	The ratio of stress to strain, within the elastic limit	Y = stress / strain = (F / A) / (x / l); unit Pa

Young's modulus is a property of the *material* itself, not of the sample's size. Steel has a much larger Young's modulus than rubber, so steel is the more elastic material in the physics sense - for the same stress it produces far less strain.

Solved Numerical 6 - force constant of a wire (textbook, page 196)

Question. When a weight of 30 N is hung from a wire of original length 2.0 m, its new length becomes 2.20 m. Calculate the force constant for the wire, if the elastic limit is not exceeded.

Solution. Extension x = 2.20 - 2.0 = 0.20 m. From F = kx, k = F / x = 30 N / 0.20 m = **150 N m^{-1}**.

Exercise MCQs - Choose the Best Answer (Textbook, page 194)

1. A container having volume of 6 m^3 is full with a liquid having density of 30 kg m^{-3}. The mass of the liquid is: A. 180 kg B. 24 kg C. 5 kg D. 0.2 kg
2. Liquids and gases are collectively categorised as: A. Liquids B. Pascals C. Fluids D. None
3. Which of the following cannot be used to measure pressure: A. Atm B. Pa C. bar D. kg m^{-3}
4. Pressure at depth in fluid: A. increases B. decreases C. none D. stays the same
5. The unit used for pressure in weather maps is: A. Atm B. Pa C. bar D. N m^{-2}
6. A rock weighs 25.7 N in air and 21.8 N in water. What is the buoyant force of the water: A. 4.1 N B. 3.9 N C. 1.18 N D. 0.84 N
7. Which of the following objects, submerged in water, experiences the largest magnitude of buoyant force: A. 1-kg helium balloon B. 1 kg of wood C. 1 kg of iron D. all the same
8. The unit of strain is: A. kg m^{-3} B. Pa C. N m^{-2} D. none
9. Young's modulus is measured in units of: A. kg m^{-3} B. Pa C. N D. none

Answer key: 1-A 2-C 3-D 4-A 5-C 6-B 7-A 8-D 9-B

Why: (1) m = ρV = 180 kg. (3) kg m^{-3} is the unit of density. (6) 25.7 - 21.8 = 3.9 N. (7) The helium balloon has the largest volume for 1 kg, so it displaces the most water. (8) Strain is a ratio of lengths. (9) Y = stress / strain, so its unit is the pascal.

Two more textbook numericals - answers

Numerical 1 (page 196). A rectangular glass block 30 cm by 5 cm by 10 cm weighs 37.5 N. Largest face area = 0.30 x 0.10 = 0.03 m^2, so least pressure = 37.5 / 0.03 = **1,250 Pa**. Smallest face area = 0.05 x 0.10 = 0.005 m^2, so greatest pressure = 37.5 / 0.005 = **7,500 Pa**.

Numerical 2 (page 196). Height of a water barometer: h = P / ρg = 1.013 x 10^5 / (1,000 x 9.8) = **about 10.3 m** - which is why mercury is used instead: its column is only 0.76 m.

One-page revision

Learn this	Core fact
Density	ρ = m / V; unit kg m^{-3}; water 1,000, ice 920, mercury 13,600
Pressure	P = F / A; unit pascal; smaller area, larger pressure
Atmospheric pressure	1 atm = 1.013 x 10^5 Pa; 1 bar = 1.000 x 10^5 Pa; barometer; falls with altitude
Pressure in liquids	P = ρhg; increases with depth; equal in all directions at one depth
Pascal's law	Transmitted equally in a confined fluid; hydraulic lift: $F_1/A_1 = F_2/A_2$
Archimedes' principle	Upthrust = weight of fluid displaced; floating body displaces its own weight
Elasticity and Hooke's law	F = -kx; k in N m^{-1}; elastic limit must not be crossed
Stress, strain, Young's modulus	Stress = F / A (Pa); strain = x / l (no unit); Y = stress / strain (Pa)